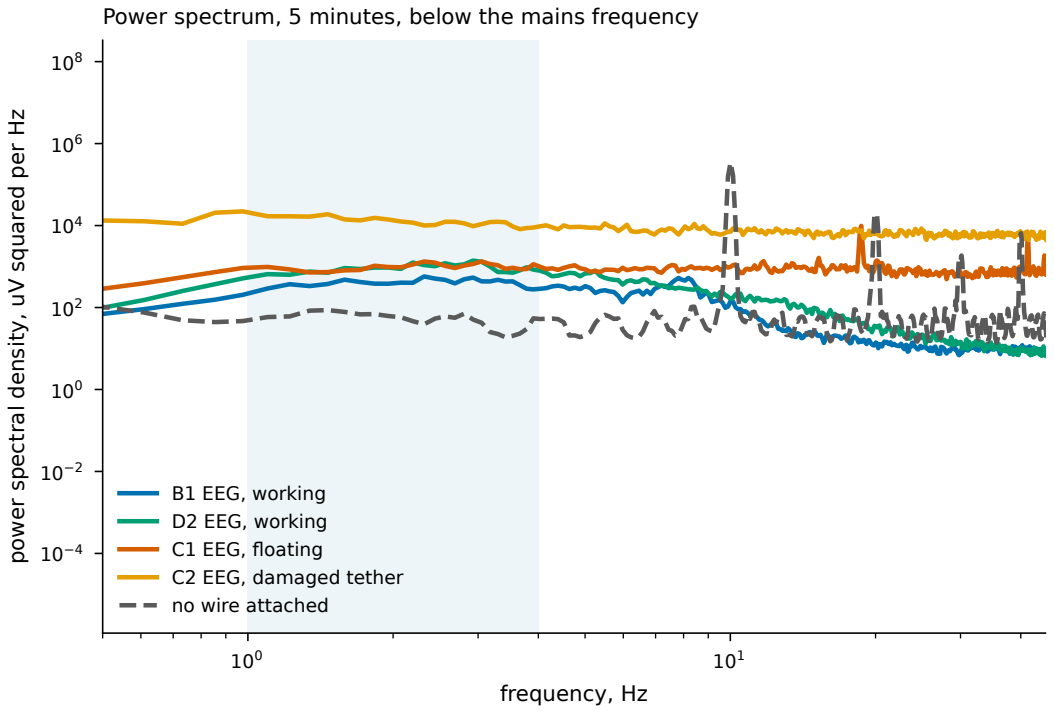
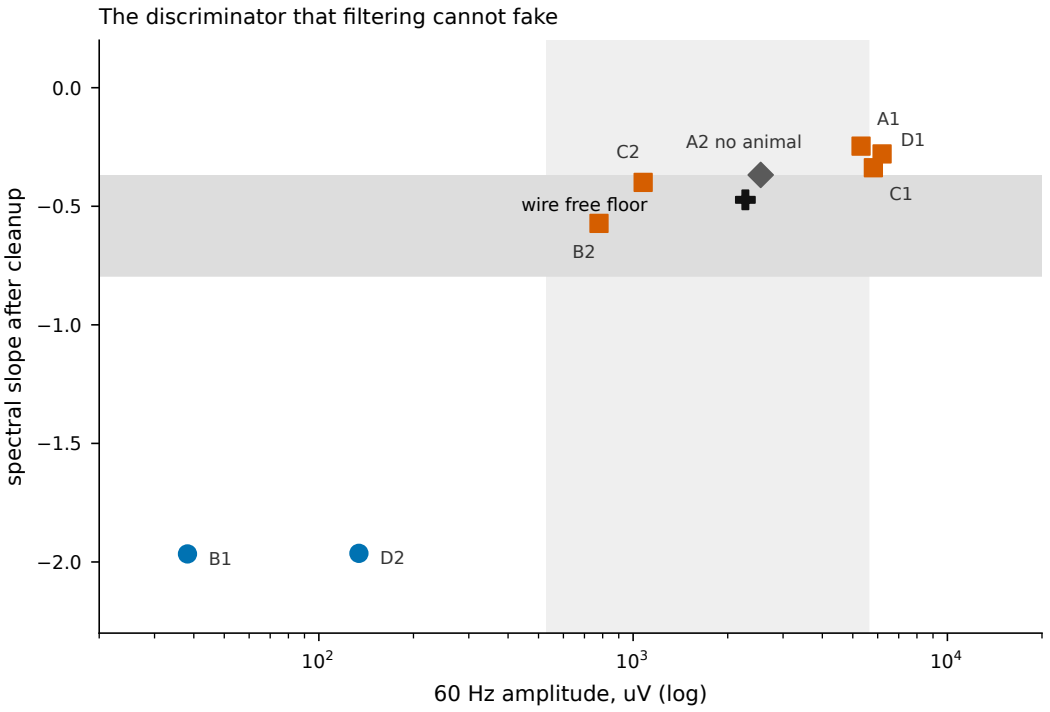
Figure 4. Spectral shape separates a real electrode from a bare wire, and mains amplitude reads out contact





LEFT. Welch power spectral density, 2048 sample segments, 5 minute windows chosen by the same median amplitude rule as figure 3, on raw signal with no filtering applied before the spectrum. A working electrode falls steeply with frequency; a floating electrode and a bare wire are close to flat. Shaded band marks the delta range of 1 to 4 Hz. RIGHT. Every point is computed by eeg-emg-analysis/qc/per_animal_forensics.py over 120 epochs sampled across the whole 23.66 hour recording, not transcribed: 60 Hz amplitude against the slope of the spectrum after notch and 0.5 to 30 Hz band pass. Grey bands are the range covered by 6 wire free inputs on both axes. The two working animals sit outside that range on both axes at once, and the two measures are independent of one another.